

Final Report

The goal of this research is to test the hypothesis that countries with populations skewed toward middle-aged or older individuals tend to have lower fertility rates and fewer births, suggesting potential population stagnation. The analysis shows that as the median age of a population increases, the total fertility rate (TFR) generally decreases, indicating a negative correlation. Similarly, when examining the dominant age group of populations, countries with older dominant age groups tend to exhibit lower TFRs, as illustrated by the histograms.

These visualizations—the scatter plot of median age versus TFR and the histogram of TFR by dominant age group—highlight the relationship between population age structure and fertility. They are relevant to the proposed trends because they clearly demonstrate how an aging population is associated with reduced fertility rates.

The findings suggest that countries with older populations are likely to face lower fertility rates and potential population stagnation. This information is useful to policymakers, as it informs them about the relationship between population age structure and fertility patterns. This relationship is relevant to workforce and economic planning, family support and healthcare planning.

Correlation Matrix

	median_age	tfr	births	pct_0_14	pct_15_19	pct_20_39	pct_40_64	pct_65_plus
median_age	1	-0.77174	-0.14469	-0.94957	-0.95304	-0.74745	0.783259	0.959655
tfr	-0.77174	1	0.087705	0.906128	0.73804	0.259938	-0.84528	-0.61878
births	-0.14469	0.087705	1	0.129151	0.147136	0.113475	-0.09918	-0.14366
pct_0_14	-0.94957	0.906128	0.129151	1	0.927254	0.536613	-0.89802	-0.83953
pct_15_19	-0.95304	0.73804	0.147136	0.927254	1	0.636273	-0.80756	-0.88472
pct_20_39	-0.74745	0.259938	0.113475	0.536613	0.636273	1	-0.38803	-0.85633
pct_40_64	0.783259	-0.84528	-0.09918	-0.89802	-0.80756	-0.38803	1	0.596959
pct_65_plus	0.959655	-0.61878	-0.14366	-0.83953	-0.88472	-0.85633	0.596959	1

Based on peer feedback, instead of using a bar chart to show TFR by predominant age group, it was suggested to use a histogram, since histograms naturally group data into bins and better represent distributions.

References

- U.S. Census Bureau. (2025). *International Database (IDB)*.
<https://www.census.gov/programs-surveys/international-programs/about/idb.html>
- Arturictus. (n.d.). *ISO 3166 Countries with Regional Codes*. GitHub Repository.
<https://github.com/luke/ISO-3166-Countries-with-Regional-Codes>